



**TA-3 Activity
of
Time Series Analysis
[RCP23DCPE611]**

Submitted by

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Abstract

Time Series Analysis (TSA) is an important technique used for forecasting future values based on historical data. In this project, electric production data is analyzed using different time series forecasting methods such as Auto Regression (AR), Moving Average (MA), ARIMA, SARIMA, and Holt-Winters Exponential Smoothing. The dataset contains monthly electric production values over a period of time. The main objective of this project is to identify trends, seasonality, and stationarity in the dataset and compare different forecasting models using evaluation metrics like RMSE, MAE, AIC, and BIC.

The project begins with loading and preprocessing the dataset followed by visualization of production patterns. Seasonal decomposition is applied to identify trend and seasonal components. Stationarity testing is performed using the Augmented Dickey-Fuller (ADF) test, and differencing is applied to make the series stationary. Various forecasting models are implemented and compared to determine the best-performing model.

The results show that seasonal models such as SARIMA and Holt-Winters perform better for electric production forecasting due to the presence of strong seasonality in the dataset. This project demonstrates the importance of time series analysis in energy forecasting and highlights how machine learning and statistical methods can assist in decision-making.

Chapter 1: Introduction

1.1 Overview of Time Series Analysis

Time Series Analysis is a statistical method used to analyze data collected over time intervals. It helps in understanding patterns such as trends, seasonality, cyclic behavior, and irregular variations. Time series forecasting is widely used in business, finance, weather forecasting, stock market analysis, healthcare, and energy production.

Electricity production forecasting is one of the most important applications of time series analysis. Accurate forecasting helps power generation companies manage resources efficiently, reduce energy wastage, and meet consumer demands.

In this project, historical electric production data is analyzed using statistical forecasting models. The forecasting models are trained on past production values and used to predict future production.

1.2 Importance of Electric Production Forecasting

- Electric production forecasting is essential because:
- It helps balance electricity supply and demand.
- It reduces operational costs.
- It supports energy planning and management.
- It prevents power shortages and overproduction.
- It improves decision-making in energy industries.

Accurate forecasting enables energy companies to allocate resources efficiently and maintain stable electricity distribution.

1.3 Objectives of the Project

The main objectives of this project are:

1. To preprocess and visualize electric production data.
2. To analyze trends and seasonal patterns in the dataset.
3. To perform stationarity testing using the ADF test.
4. To apply differencing to make the series stationary.
5. To implement forecasting models such as AR, MA, ARIMA, SARIMA, and Holt-Winters.

6. To compare models using RMSE, MAE, AIC, and BIC.
7. To identify the best forecasting model.

1.4 Scope of the Project

This project focuses on monthly electric production forecasting using statistical time series techniques. The project can be extended in the future using deep learning methods such as RNN, LSTM, and GRU for more accurate forecasting.

Chapter 2: Dataset Description

2.1 Dataset Overview

The dataset used in this project contains historical electric production values recorded monthly. The dataset includes:

DATE: Represents the month and year.

Production: Represents electric production values.

The dataset is converted into a time-indexed format to perform time series analysis.

2.2 Data Preprocessing

Data preprocessing is an important step in machine learning and time series analysis. The following preprocessing steps are performed:

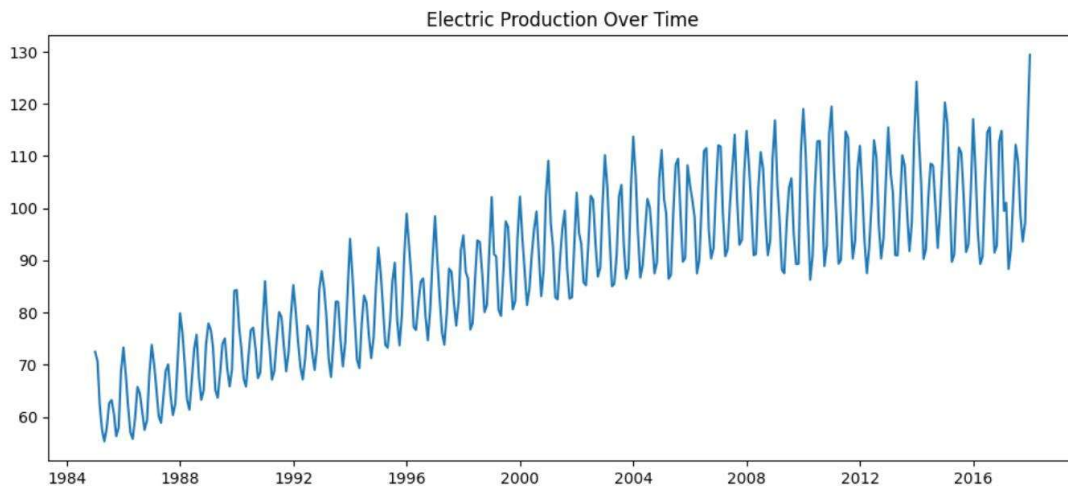
1. Importing required libraries.
2. Loading the CSV dataset.
3. Renaming columns.
4. Converting date columns into datetime format.
5. Setting the date column as index.
6. Checking for missing values.
7. Sorting data chronologically.

2.3 Data Visualization

Visualization helps understand trends and seasonal behavior in the dataset.

Production Plot

A line graph is plotted to observe electric production changes over time.



The graph shows fluctuations in production values and indicates the presence of trend and seasonality.

Chapter 3: Seasonality Analysis

3.1 What is Seasonality?

Seasonality refers to repetitive patterns occurring at regular intervals in time series data. In electric production datasets, seasonal variations may occur due to weather conditions, industrial usage, and consumer demand.

3.2 Seasonal Decomposition

Seasonal decomposition separates the time series into:

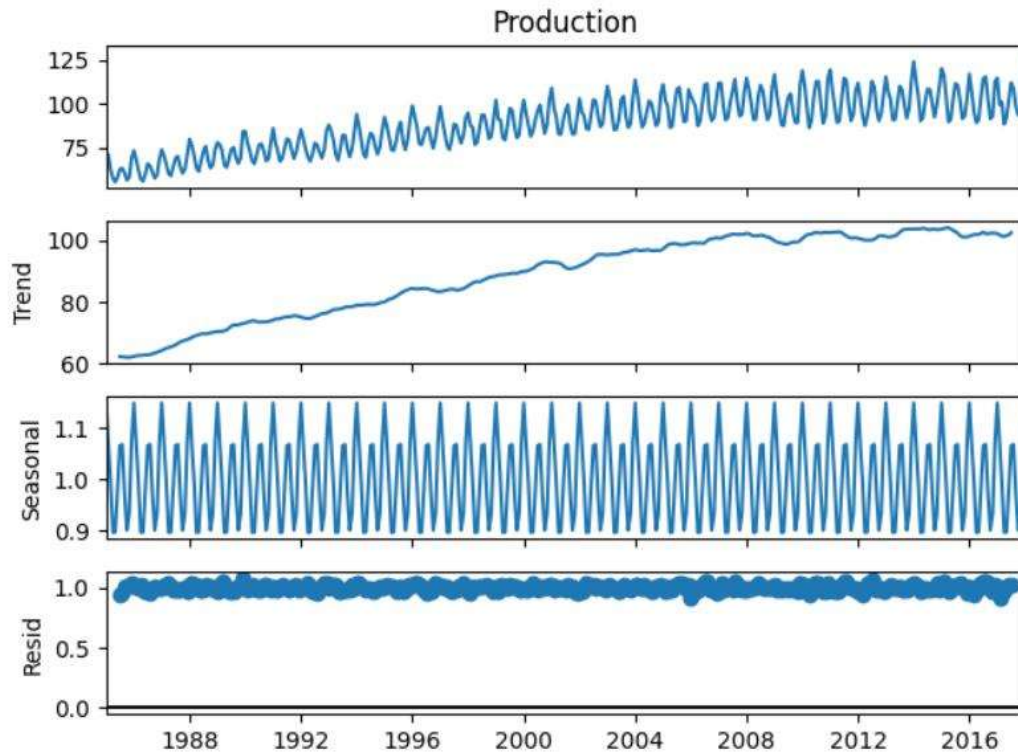
Trend Component

Seasonal Component

Residual Component

Code for Seasonal Decomposition

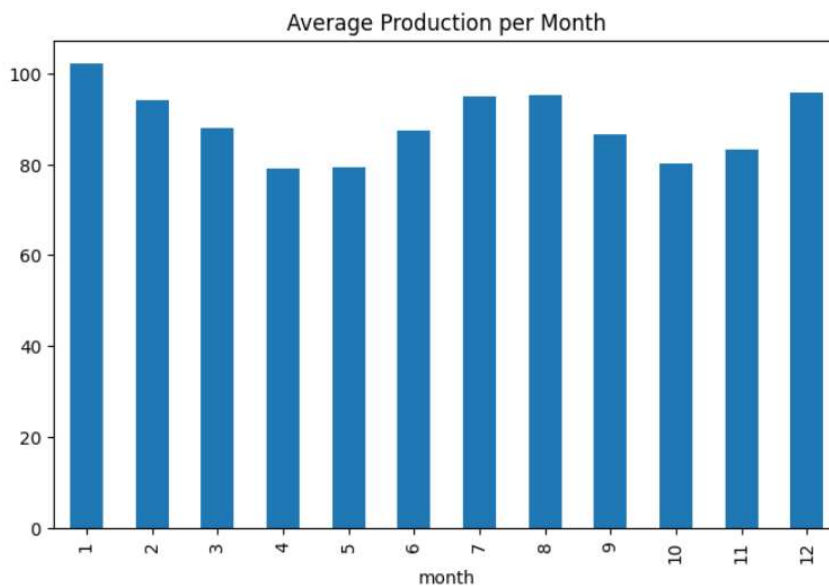
```
from statsmodels.tsa.seasonal import seasonal_decompose
decomp = seasonal_decompose(df['Production'], model='multiplicative')
decomp.plot()
plt.show()
```



The decomposition graph clearly shows recurring seasonal patterns.

3.3 Monthly Pattern Analysis

Monthly average production values are calculated to identify seasonal fluctuations.



The bar graph helps identify months with high and low production.

Chapter 4: Stationarity Check

4.1 What is Stationarity?

A time series is said to be stationary when its statistical properties such as mean and variance remain constant over time.

Stationarity is important because many forecasting models assume the data is stationary.

4.2 Augmented Dickey-Fuller (ADF) Test

The ADF test is used to check whether the series is stationary.

Hypothesis of ADF Test

Null Hypothesis (H0): Data is non-stationary.

Alternative Hypothesis (H1): Data is stationary.

Interpretation

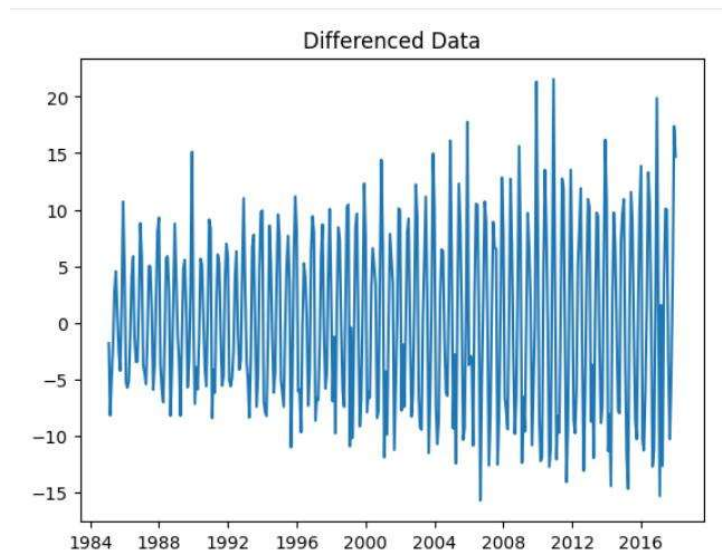
If $p\text{-value} > 0.05 \rightarrow$ Data is non-stationary.

If $p\text{-value} < 0.05 \rightarrow$ Data is stationary.

The electric production dataset initially shows non-stationary behavior.

4.3 Differencing

Differencing is applied to remove trend and make the series stationary.



After differencing, the dataset becomes more stable.

Chapter 5: Train-Test Splitting

5.1 Importance of Train-Test Split

The dataset is divided into training and testing data:

Training Data: Used to train forecasting models.

Testing Data: Used to evaluate model performance.

Code

```
train = df.iloc[:-12]
```

```
test = df.iloc[-12:]
```

The last 12 months are reserved for testing.

Chapter 6: Forecasting Models

6.1 Auto Regression (AR)

The Auto Regression model predicts future values using previous observations.

Features of AR Model

Uses lag values.

Suitable for stationary data.

Captures linear relationships.

Advantages

Simple and efficient.

Good for short-term forecasting.

Limitations

Cannot capture seasonality effectively.

Sensitive to non-stationary data.

6.2 Moving Average (MA)

The Moving Average method predicts future values based on the average of past observations.

Advantages

Easy to implement.

Reduces noise.

Limitations

Poor long-term forecasting.

Cannot handle complex patterns.

6.3 ARIMA & SARIMA Model

ARIMA stands for:

AR → Auto Regression

I → Integrated

MA → Moving Average

ARIMA combines autoregression, differencing, and moving averages.

Components of ARIMA

p → Number of autoregressive terms.

d → Number of differencing operations.

q → Number of moving average terms.

Advantages

Handles trend and stationarity.

Widely used forecasting model.

Limitations

Not ideal for seasonal data.

Parameter tuning is required.

SARIMA stands for Seasonal ARIMA. It extends ARIMA by handling seasonality.

Components of SARIMA

Seasonal autoregression

Seasonal differencing

Seasonal moving average

Advantages

Handles trend and seasonality.

Provides accurate forecasting.

Limitations

Computationally expensive.

Requires parameter tuning.

6.5 Holt-Winters Exponential Smoothing

Holt-Winters forecasting uses exponential smoothing techniques for trend and seasonality.

Types of Exponential Smoothing

1. Simple Exponential Smoothing
2. Holt's Linear Trend Method
3. Holt-Winters Seasonal Method

```
from statsmodels.tsa.holtwinters import ExponentialSmoothing

model_hw = ExponentialSmoothing(train['Production'],
                                seasonal='multiplicative',
                                seasonal_periods=12).fit()

forecast_hw = model_hw.forecast(len(test))
```

Advantages

Excellent for seasonal datasets.

Fast and effective.

Limitations

Sensitive to outliers.

Works best with stable seasonality.

Chapter 7: Model Evaluation

7.1 AIC and BIC Comparison

AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) are used for model selection.

```
print("AR AIC:", model_ar.aic)
print("ARIMA AIC:", model_arima.aic)
print("SARIMA AIC:", model_sarima.aic)
print("Holt-Winters AIC:", model_hw.aic)
print('-----')
print("AR BIC:", model_ar.bic)
print("ARIMA BIC:", model_arima.bic)
print("SARIMA BIC:", model_sarima.bic)
print("Holt-Winters BIC:", model_hw.bic)
```

```
AR AIC: 1816.8259085613295
ARIMA AIC: 2011.4541640073824
SARIMA AIC: 1702.0190819498216
Holt-Winters AIC: 690.7699715592963
-----
AR BIC: 1871.7280064363429
ARIMA BIC: 2031.207376770321
SARIMA BIC: 1721.6135512211874
Holt-Winters BIC: 746.1153782393253
```

Lower AIC and BIC values indicate better models.

7.2 RMSE and MAE

RMSE (Root Mean Square Error)

Measures prediction error magnitude.

MAE (Mean Absolute Error)

Measures average absolute prediction error.

```
from sklearn.metrics import mean_squared_error
rmse_ma = np.sqrt(mean_squared_error(test['Production'], forecast_ma))
rmse_ar = np.sqrt(mean_squared_error(test['Production'], forecast_ar))
rmse_arima = np.sqrt(mean_squared_error(test['Production'], forecast_arima))
rmse_sarima = np.sqrt(mean_squared_error(test['Production'], forecast_sarima))
rmse_hw = np.sqrt(mean_squared_error(test['Production'], forecast_hw))

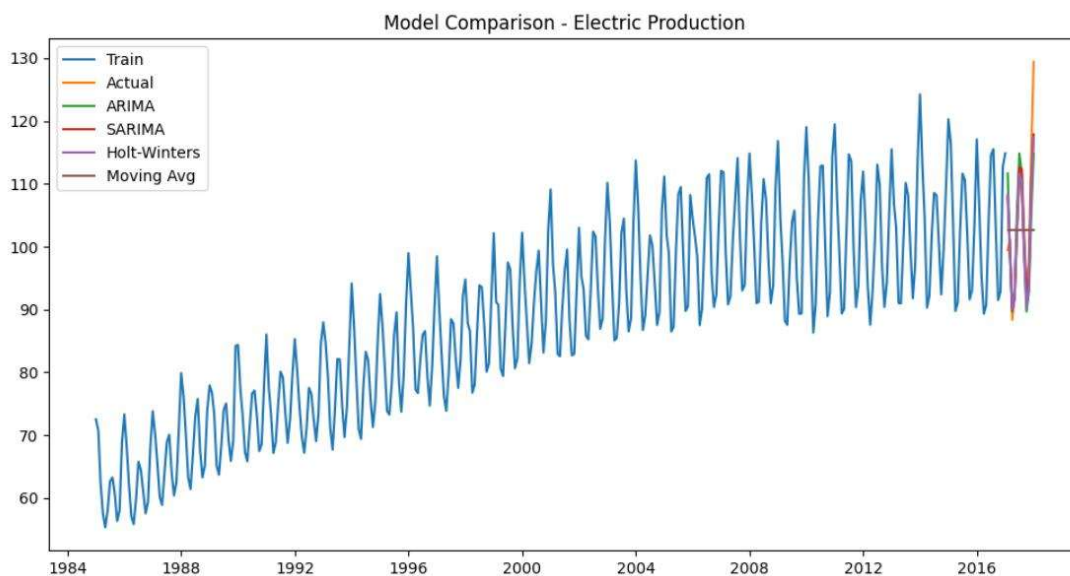
print("AR RMSE:", rmse_ar)
print("Moving Avg:", rmse_ma)
print("ARIMA:", rmse_arima)
print("SARIMA:", rmse_sarima)
print("Holt-Winters:", rmse_hw)
print("Moving Avg:", rmse_ma)
```

```
AR RMSE: 5.4524412197164605
Moving Avg: 10.982412637817415
ARIMA: 6.556317211801126
SARIMA: 4.6161851878145175
Holt-Winters: 4.969390680878857
Moving Avg: 10.982412637817415
```

Lower RMSE and MAE values indicate better forecasting performance.

Chapter 8: Visualization of Forecasting Results

Visualization is used to compare actual and predicted values.



The graph helps visualize forecasting accuracy.

Chapter 9: Best Model Selection

Different forecasting models are compared using evaluation metrics.

Observations

AR and MA models provide basic forecasting.

ARIMA improves performance using differencing.

SARIMA performs better due to seasonality handling.

Holt-Winters also gives strong performance.

SARIMA is selected as the best model because it captures both trend and seasonal variations.

Chapter 10: Applications of Time Series Forecasting

Time series forecasting has applications in multiple industries:

Energy Sector

Electricity demand forecasting

Load balancing

Resource allocation

Finance

Stock market prediction

Risk analysis

Revenue forecasting

Healthcare

Disease outbreak prediction

Patient admission forecasting

Retail

Sales forecasting

Demand prediction

Weather Forecasting

Temperature prediction

Rainfall forecasting

Chapter 11: Advantages and Limitations

Advantages

1. Helps identify trends and patterns.
2. Improves future prediction accuracy.
3. Supports decision-making.
4. Useful in many industries.
5. Handles seasonal data effectively.

Limitations

1. Sensitive to data quality.
2. Requires large historical datasets.
3. Complex parameter tuning.
4. Accuracy depends on stationarity.
5. Cannot always predict sudden changes.

Chapter 13: Conclusion

This project successfully demonstrates the implementation of Time Series Analysis for electric production forecasting. Different forecasting techniques such as Auto Regression, Moving Average, ARIMA, SARIMA, and Holt-Winters Exponential Smoothing were applied to historical electric production data.

The dataset was first preprocessed and visualized to understand trends and seasonality. Seasonal decomposition confirmed the presence of recurring patterns in electric production. The Augmented Dickey-Fuller test revealed that the data was non-stationary, and differencing was applied to stabilize the series.

Among all forecasting methods, SARIMA and Holt-Winters performed the best because they effectively captured seasonal variations. Evaluation metrics such as RMSE, MAE, AIC, and BIC were used to compare the models.

The project highlights the importance of time series forecasting in the energy sector. Accurate forecasting helps organizations optimize electricity generation, reduce costs, and improve resource management.

Future work can include advanced deep learning models such as LSTM, GRU, and hybrid forecasting methods for improved prediction accuracy.